CODECHEF

The CodeChef.com Challenge—your monthly dose of coding puzzles from India's biggest coding contest!

Solve this month's puzzle, and you stand the chance of being one of the three lucky people to win a cash prize of Rs 1,000 each!

The November edition's puzzle::

Our chef is fond of collecting international and national postage stamps, but only he knew how many stamps he owned. Once, when asked by one of his friends about the number of stamps he had, he replied with the following sentence:

"If I divide the stamps into two sets—international and national—then five times the difference between the number of stamps in each set equals half of the total number of stamps. Also, I've more national stamps than international ones. And half of the difference between the squares of the number of stamps in the national and international sets is 1620."

This answer left his friend rather puzzled. Can you help the chef's friend to find out how many international and national stamps he has collected?

The solution

Let the number of Indian and international stamps be 'x' and 'y'.

Given,

Condition 1: 5(x-y) = (x+y)/2

(x+y)/(x-y) = 10

Condition 2: (x^2-y^2)/2 = 1620

(x+y)(x-y) = 3240

Let (x+y) = A and (x-y) = B;

=> A/B = 10 ; A = 10B and AB = 3240

Therefore, by solving the equations, we get, A = 180 and B = 18;

On putting values,

(x+y) = 180 and (x-y) = 18

On further solving, we get,

x = 99 and y = 81.

ANSWER:

The number of Indian stamps is: 99

The number of international stamps is: 81

And the winners this month are:

- Santosh Pathak
- Vishal Gawai
- Vijay Kumar

Here's your 'CodeChef Puzzle of the Month':

Our chef is fond of playing games. He called a few children to his restaurant to play a 'smart guessing game'.

In this game, the player will use N coins numbered from 1 to N, and all the coins will be showing heads.

The player needs to play N rounds. In the j-th round, the player will flip the face of all the coins whose number is less than or equal to j (i.e., the face of coin i will be reversed, from Heads to Tails, or from Tails to Heads, for i ≤ j).

Each of the children has to guess the total number of coins showing Heads and Tails, respectively, after playing N rounds. Consider N to be 500 and help these children in finding out the answer to the chef's question.

Looking for a little extra cash?

Solve this month's CodeChef Challenge and send in your answer to *codechef@efyindia.com* or *lfy@codechef.com* by December 15, 2012. Three lucky winners get to win some awesome prizes both in cash and other merchandise! You can also participate in the contest by visiting the online space for the article at *https://www.facebook.com/LinuxForYou* END

About CodeChef: CodeChef.com is India's first, non-commercial, online programming competition, featuring monthly contests in more than 35 different programming languages. Log on to CodeChef.com for the December Challenge that takes place from the 1st to the 11th, to win cash prizes up to Rs 20,000. Do keep visiting CodeChef to participate in the multiple programming contests that take place on the website throughout the month!